MGU-UGP (HONOURS) REGULAR EXAMINATION APRIL 2026

SECOND SEMESTER

Multi-Disciplinary Courses (MDC) - MG2MDCGEO100 - NATURAL HAZARDS (2024 ADMISSION ONWARDS)

Duration: 1 Hours **Maximum Marks: 35**

QP Code-25901690

ANSWER SCHEME

Part A

Multiple choice /Answer all question/ Each question carries 1 marks [1x6 = 6]

1. b) Underwater earthquakes
2. a) Too much rain and water
3. b) Indian Meteorological Department (IMD)
4. c) Flood and Landslide
5. b) Dengue
6. a) A large ocean wave caused by an earthquake or volcanic eruption

Part B

Short answer / Answer any 5 questions /Each question carries 3 marks [3x5 = 15]

7. Earthquake Risk Factors

Population Density – More people in an area means more damage and deaths.

Building Quality – Weak or old buildings collapse easily during earthquakes.

Emergency Preparedness – Good rescue systems and public awareness reduce loss of life.

Tectonic Activity – Places near fault zones are more likely to experience earthquakes.

Ground Conditions – Soft soil can increase shaking and cause liquefaction.

Secondary Hazards – Earthquakes can also cause landslides, tsunamis, fires, floods, and aftershocks.

8. Health risks associated with animals to human

Spread of diseases – Animals can spread diseases like rabies and bird flu to humans through bites, food, water, or contact.

Allergies and breathing problems – Animal fur, feathers, and waste can cause allergies and asthma.

Injuries and poisoning – Some animals may bite, sting, or inject venom, causing injury or sickness.

9. Role of dams and levees in mitigating flood risks through engineering solutions

Dams hold extra water and release it slowly to reduce floods.

Levees are walls or embankments along rivers that stop water from overflowing.

Both help protect people, houses, and land from flood damage.

10. Importance of understanding natural hazards.

It helps people prepare and take safety measures before disasters like floods, earthquakes, landslide, Tsunami,cyclones,Volcanism.. etc

It reduces loss of life and property through proper planning, warning systems, and disaster management.

It helps governments and communities develop safer buildings and better emergency response plans.

11. Natural hazards destroy habitats, so plants and animals lose their homes.

Many plants and animals die during disasters.

Disasters disturb food chains and can cause some species to disappear.

12. Cyclones form over warm oceans where the air becomes warm and moist.

The warm air rises, creating a low-pressure area, and more air moves in.

As the air rises, it cools and condenses, forming clouds and releasing heat, which makes the system stronger.

The Coriolis Effect (due to Earth's rotation) causes the winds to spin around the low-pressure centre, forming a cyclone.

13. Community-based early warning systems are important because they help detect hazards early and warn people before a disaster happens, which reduces loss of life and damage. Since local people are involved, warnings spread quickly and responses are faster. They also increase awareness and help people stay prepared, making evacuation and safety actions easier during emergencies.

Part C

Short note / Answer any 2 questions /Each question carries 7 marks [7x2 = 14]

14.Geological hazards: Caused by internal Earth processes, especially tectonic plate movements.

Examples: Earthquakes / Volcano eruption /Landslides /Tsunamis

Meteorological hazards: Caused by atmospheric conditions such as temperature, pressure, wind, and moisture. Cyclone / Flood /Droughts /Heatwaves/Thunderstorms.

Describe with examples

15. Mining and Quarrying /Deforestation / Climate Change due to Human Activities /Construction in Hazard-prone Areas /Urbanizatio /Over-extraction of Groundwater /Industrialization. Briefly explain each

16. List out the major natural hazards: Earthquakes/Landslides /Tsunami / Cyclones

Floods

17. Flood Preparedness : Flood forecasting and early warning systems /Flood hazard mapping /Public awareness and education /Proper drainage system maintenance /Emergency planning and evacuation strategies /Avoiding construction in flood

Flood Mitigation : Structural Measures /Non-Structural Measures /Environmental Measures

Briefly explain each